

BEHAVIORAL INTERVIEW — PROJECT STORY MATRIX

Print-friendly A4 landscape | Use the project stories below as your behavioral interview story bank.

Common Questions	PROJECT 1 — INTELLIGENT GROUP TRAVEL PLANNER	PROJECT 2 — CONTROL SYSTEMS SIMULATOR	PROJECT 3 — RESEARCH PAPER ANALYSIS PLATFORM
Challenges	Designing a multi-step planning flow while keeping trip details, members, budgets and preferences consistent. Handling variable AI-generated output.	Turning Mason's Gain Formula into a usable web tool while correctly handling multiple forward paths, loops and non-touching loop combinations.	Building a reliable RAG pipeline combining dense retrieval with BM25 while keeping answers grounded in uploaded research papers.
Mistakes / Failures	Initially focused on itinerary generation before making validation robust. Incomplete/conflicting input could lead to inconsistent AI output. Learned to validate and structure data first.	Initially focused on the mathematical computation and underestimated unusual graph edge cases. Learned to validate against simple cases before complex structures.	Initially assumed semantic similarity alone would be enough. Testing showed BM25 was useful for exact technical terminology, leading to hybrid retrieval.
Enjoyed	Integrating the AI layer with application logic and seeing structured preferences become personalized itineraries.	Translating a mathematical/control-systems concept into an interactive tool users could experiment with.	Understanding retrieval behavior and combining different retrieval strategies to improve the quality of results.
Leadership	Took ownership of the application flow and broke it into trip input, member management, validation and itinerary-generation components.	Took ownership of the complete implementation, from understanding Mason's Gain Formula to implementing the computation and exposing it through Flask.	Took ownership of the RAG architecture, deciding how ingestion, indexing, retrieval and QA should interact and how FAISS and BM25 complement each other.
Conflicts	Different priorities: add more AI features vs. stabilize the core workflow. I pushed for reliability first because feature-rich output is not useful with inconsistent input.	Trade-off between functionality and UI/feature additions. I preferred validating the core computation first because incorrect results make extra features meaningless.	Trade-off between simpler semantic-only search and hybrid retrieval. I preferred testing retrieval quality before selecting the simpler architecture.
What You'd Do Differently	Define the data model and validation rules earlier and establish clearer interfaces between frontend, backend and AI layers.	Build a larger automated mathematical test suite earlier, especially for multiple loops and non-touching-loop edge cases.	Create a formal evaluation dataset and retrieval metrics earlier instead of relying mainly on manual testing.

YOUR 3 MAIN STORIES

1. PLANORA — Main Behavioral Project

Use for: challenging project, failure, conflict, leadership, problem-solving, what you would do differently.

STAR answer: While building an AI-powered group travel planner, one challenge was maintaining consistency across multiple steps. Users could enter trip details, add members, specify budgets and preferences, and finally generate an itinerary. I realized that if we didn't validate and structure the data before sending it to the AI layer, the output could become inconsistent. I therefore made validation a separate part of the workflow, checking dates, budget, member information and preferences before planning. I also structured the AI interaction so the model received consistent inputs rather than raw user data. **Lesson:** AI doesn't replace good application design; output quality depends heavily on input quality and consistency.

2. MASONYX — Technical Problem-Solving Story

Use for: most technically challenging project, difficult problem, debugging, failure, something built from scratch.

STAR answer: Masonyx was challenging because the core problem was not standard CRUD work. I had to implement Mason's Gain Formula, involving forward paths, individual loops and combinations of non-touching loops. I broke the problem into smaller components instead of calculating the final transfer function directly. I separately identified paths and loops, calculated their gains, and then used those results to construct the final expression. I tested simpler graphs before moving to more complex cases. **What I'd change:** build a larger automated test suite earlier, especially for multiple-loop edge cases.

3. RESEARCH PAPER ANALYZER — Technical Depth Story

Use for: architecture, technical decisions, trade-offs, RAG, retrieval quality, optimization.

STAR answer: One major challenge was retrieval quality. Initially, semantic embeddings seemed sufficient for natural-language questions, but testing showed that research queries often contain exact terms, abbreviations and specific concepts. I therefore combined dense retrieval using FAISS with BM25 lexical retrieval. FAISS captured semantic similarity while BM25 helped with exact terminology. This hybrid approach was more robust than relying on either method alone. **Lesson:** choosing a technology is not enough; you need to understand its failure modes and design around them.

BEHAVIORAL STORY BANK — MEMORIZE THESE 6

Story	Project	Questions it covers
AI input validation problem	Planora	Challenge, failure, problem-solving
AI output reliability	Planora	Failure, learning, improvement
Mason's Formula implementation	Masonryx	Technical challenge
Graph edge cases	Masonryx	Debugging, failure
Semantic vs keyword retrieval	Research Analyzer	Technical decision, trade-off
Hybrid FAISS + BM25	Research Analyzer	Architecture, optimization

INTERVIEW STRATEGY

Don't memorize 30 separate answers. Memorize these six stories and adapt each one to the question. For conflict questions, don't invent dramatic arguments. Explain the differing viewpoints, the trade-off, your reasoning, and how the team reached a decision.

NIC: Your resume also includes a Software Development Intern role at NIC and collaboration with cross-functional teams. Prepare one separate NIC story for teamwork/leadership; it can be stronger than forcing every behavioral question onto a project.